NKUMBA UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

DESIGN AND IMPLEMENTATION OF A WEB-BASED LOST AND FOUND PROPERTY MANAGEMENT SYSTEM WITH AI-ASSISTED ANALYTICS: A CASE STUDY OF NKUMBA UNIVERSITY

BY

[STUDENT NAME(S)]

[REGISTRATION NUMBER(S)]

Supervisor: [SUPERVISOR NAME]

A research proposal submitted to the School of Computing and Informatics in partial fulfilment of the requirements for the award of a Bachelor's degree of Nkumba University.

[MONTH, YEAR]

Contents

CHAPTER ONE: INTRODUCTION — 1.1 Background to the Study; 1.2 Statement of the Problem; 1.3 Objectives; 1.4 Research Questions; 1.5 Scope of the Study; 1.6 Justification; 1.7 Significance; 1.8 Definition of Key Terms; 1.9 Assumptions and Constraints

CHAPTER TWO: LITERATURE REVIEW — 2.1 Introduction; 2.2 Review Approach; 2.3 Lost-and-Found Procedures and Custody; 2.4 Web-Based Item Registers; 2.5 Claims, Ownership Verification and Audit Trails; 2.6 AI-Assisted Matching and Descriptive Analytics; 2.7 Data Protection and Security; 2.8 Related Projects; 2.9 Research Gap; 2.10 Conceptual Framework; 2.11 Chapter Summary

CHAPTER THREE: METHODOLOGY — 3.1 Introduction; 3.2 Research Design; 3.3 Study Site; 3.4 Study Population; 3.5 Sample and Sampling; 3.6 Data Collection Instruments; 3.7 Data Collection Procedures; 3.8 System Requirements and Workflow Specification; 3.9 AI Analytics Boundary; 3.10 Privacy and Security Requirements; 3.11 Architecture, Modules and Development Process; 3.12 Evaluation Design; 3.13 Data Analysis; 3.14 Quality Assurance; 3.15 Ethical Considerations; 3.16 Limitations; 3.17 Chapter Summary

REFERENCES

APPENDICES — I Student Questionnaire; II Custodian and Staff Interview Guide; III Observation Checklist; IV Indicative Work Plan; V Indicative Budget; VI Register of Open Items to Confirm

Note: page numbers and the formal table of contents are generated after final formatting in the exported Word file.

CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

People lose and find personal property wherever many of them share space. Identification cards, phones, keys, books, bags, chargers and clothing are misplaced in lecture rooms, libraries, laboratories and corridors. A university organised into several schools and many buildings multiplies the places where an item can be left behind and, equally, the places where it can be handed in. Someone who finds an item may not know who owns it or which office should receive it, and someone who has lost an item may not know whether it has been found or where it is being kept.

Where no shared record exists, this information travels through notice boards, messaging groups, word of mouth and visits to offices. These channels work for small volumes but scatter information. A Makerere University project on lost identification cards describes this pattern: found cards were placed at different receptions around the university, so a student searching for a card had no way of knowing where it might be (Mirembe et al., 2023). Nkumba University operates through seven constituent schools (Nkumba University, n.d.-a), so a found item reported to one school's office may never be seen by an owner who belongs to another school. How lost and found property is actually handled at Nkumba is not yet documented for this study and will be established through requirements research (Chapter Three).

A web-based system can give lost and found property one searchable campus-wide record. It also introduces design problems that a simple notice board does not have. First, a report that an item has been found is not the same thing as the University holding the item: a finder may post a report and then fail to hand the item in, or hand it in to an office the system does not know about. The system must therefore distinguish a published report from confirmed physical custody. Second, listings must help owners recognise their property without revealing the details that prove ownership; otherwise a false claimant can copy the description. Third, student identities, contact details and claim evidence are personal data and must be visible only to authorised staff who need them. Fourth, decisions to release property need a traceable record of who decided what and why.

This study therefore proposes the design, implementation and evaluation of a web-based Lost and Found Property Management System with AI-assisted analytics for Nkumba University. Students create accounts with their student details in order to post lost or found reports and submit claims. Authenticated university users can browse and search approved listings by category. A student who finds an item posts a report and then delivers the item to the School Administrator or other designated custodian of the relevant school, who alone confirms receipt into custody. Custodians review claims, record their decisions and authorise handover under a procedure the University approves. Administrators may edit or hide listings, but hiding is a soft delete that retains the record and its audit trail.

The AI-assisted component has a deliberately narrow role. Text descriptions of lost and found items can be represented using term frequency–inverse document frequency (TF-IDF) and compared using cosine similarity, a standard information-retrieval approach that is documented in the scikit-learn library (Scikit-learn developers, n.d.-a, n.d.-b; Manning et al., 2008). The system will use such scores to suggest ranked candidate matches for human review and to produce descriptive analytics, such as counts by category, school, period and status. A similarity score is a retrieval aid. It is never evidence of ownership and never authorises release.

The project is an applied computing study that produces a working software artefact for an identified information-management problem. Design science research is suited to such work because it creates and evaluates artefacts that address practical problems (Hevner et al., 2004), and the process model of Peffers et al. (2007) provides stages from problem identification to communication of results. These guide the structure of the study.

1.2 Statement of the Problem

Lost and found property at a multi-school university presents an information-management problem rather than simply an absence of software. Students need to discover, across the whole campus, whether an item has been reported. The physical custodian of a found item and its current status need to be clear to anyone searching. Claims on an item must be verified in a controlled way that resists false claims and leaves a trace of who decided what. Administrators need a reliable record from which they can see which items are awaiting delivery, held, claimed or returned.

If these needs are not met, owners may not learn that an item has been found, found items may sit unrecorded or in the wrong office, and informal release of property may expose the University and its students to disputes over who received what. Publishing too much detail about an item invites false claims, while collecting personal data without safeguards creates privacy risk.

The current procedures at Nkumba University have not yet been investigated for this study, and the extent of these problems is therefore a matter for requirements research rather than assertion. The study will address the problem by establishing the actual procedures and requirements and then designing, building and evaluating a prototype that centralises lost and found records, separates reporting from physical custody, controls claim verification and uses text similarity to suggest possible matches for human review.

1.3 Objectives

1.3.1 General Objective

To design, implement and evaluate a web-based Lost and Found Property Management System with AI-assisted analytics to support reporting, discovery, custody, claim verification and return of property at Nkumba University.

1.3.2 Specific Objectives

1. To investigate current lost and found reporting, custody, search, claim-verification and handover procedures at Nkumba University and derive the functional, privacy and security requirements for the system.
2. To design and implement student accounts, searchable campus listings, school-level custody, claim review, administrative moderation and auditable handover workflows.
3. To develop bounded AI-assisted candidate matching and descriptive analytics for lost and found records.
4. To evaluate the prototype for functionality, usability, security and privacy controls, auditability and the relevance of candidate matches.

1.4 Research Questions

Each research question corresponds to one specific objective.

1. What procedures are currently used to report, store, search for, verify claims on and hand over lost and found property at Nkumba University, what challenges do students and custodians report, and what functional, privacy and security requirements follow? (Objective 1)
2. How can accounts, listings, custody, claims, moderation and handover be designed so that a posted report is distinguished from physical custody and private evidence is separated from what other users see? (Objective 2)

3. How useful are TF-IDF-based candidate matching and descriptive analytics for lost and found records, and what are their limitations? (Objective 3)
4. To what extent does the prototype meet its functional, usability, security and privacy, auditability and matching-relevance criteria? (Objective 4)

Table 1.1: Link between objectives, research questions and methods

Objective	Research question	Main method (Chapter Three)	Output
1. Investigate procedures; derive requirements	RQ1	Questionnaire, interviews, observation, document review; thematic and descriptive analysis	Requirements specification
2. Design and implement the workflows	RQ2	UML and database design; iterative build; permission matrix and status model	Working prototype
3. Develop AI matching and analytics	RQ3	TF-IDF and cosine similarity; labelled test set; precision@k and recall	Ranked suggestions; dashboard
4. Evaluate the prototype	RQ4	Test scenarios; security tests; user testing; error review	Evaluation results

1.5 Scope of the Study

1.5.1 Content Scope

The study covers requirements gathering, analysis, design, implementation and evaluation of a prototype for lost and found property only. The prototype will include: student account creation with an approved identity check; browsing and category search of approved listings by authenticated university users; lost-item and found-item reports; recording of physical receipt into custody by an authorised School Administrator or custodian; custody transfers between offices; claim submission, review and decision, including more than one claim on one item; administrative editing and hiding of listings with a reason; an audit trail; handover records; candidate matching; and descriptive analytics.

Three roles are defined: verified student, School Administrator or Lost and Found Custodian, and system administrator. One person may hold more than one role, but the system enforces each role's permissions separately. Listings are visible to authenticated university users only, unless the University explicitly approves wider access.

The prototype does not identify or assess people, decide ownership, or automate the release of any item. Image recognition, multi-institution operation and any form of theft or incident reporting are outside the scope of the study.

1.5.2 Geographical Scope

The study is a case study of Nkumba University. The University's site describes Entebbe and Kampala campuses (Nkumba University, n.d.-b), so which campus or campuses the prototype serves, and which school offices accept found property, will be settled with the supervisor and the University.

1.5.3 Time Scope

The research and prototype development will take place within a period to be approved by the supervisor and aligned with the School's research schedule (Appendix IV).

1.5.4 Technical Scope

The proposed implementation uses HTML, CSS and JavaScript for a responsive browser interface, PHP for application logic, MySQL for data storage and Python with scikit-learn for similarity analysis. The final stack will be confirmed after requirements analysis. Core matching uses text and structured metadata. Image upload, if included at all in the prototype, will not be used for matching.

1.6 Justification of the Study

The study is justified by the need for an organised, traceable and privacy-respecting way of handling lost and found property across a university with several schools. A shared register can reduce dependence on scattered notices and single-office knowledge. Separating published reports from confirmed custody and keeping ownership evidence private address the practical risks of misplaced responsibility and false claims. Similarity suggestions may help custodians and students notice possible matches that are described in different words. The study is also academically justified because it applies software engineering, database design, web security and an information-retrieval technique to a real problem and evaluates the artefact using a design science approach.

1.7 Significance of the Study

To students. A single place to report property, see what has been found and follow the progress of a claim, without visiting several offices. Benefits will be evaluated, not assumed.

To School Administrators and custodians. A structured record of what has been received, where it is held, who has claimed it and what was decided, together with a defined verification workflow.

To University administrators. Auditable records and aggregate reports on categories, locations, claims and returns, subject to data quality.

To the academic community. An applied, documented example of combining a role-controlled workflow, privacy-preserving listings and explainable text matching, which may inform later student projects and research.

1.8 Definition of Key Terms

Lost-item report: a record created by a student describing property they have lost.

Found-item report: a record created by a student or finder describing property they have found. It does not mean the University has received the item.

Custody: the confirmed physical holding of an item by an authorised School Administrator or custodian, recorded with the responsible office and a restricted storage reference.

Claim: a request by a verified student to receive an item, reviewed by a custodian. Account identity shows who made the claim; it does not by itself prove ownership.

Soft delete: hiding, archiving or withdrawing a record from ordinary browsing while retaining it and its audit trail.

Candidate match: a ranked suggestion that a lost report and a found report may describe the same item, produced by text similarity and always subject to human verification.

Descriptive analytics: summaries of existing records, such as counts by category or status, calculated by the database.

1.9 Assumptions and Constraints

The study assumes that the University and the supervisor will permit requirements gathering and prototype evaluation, that participants will give honest feedback, and that suitable test data can be created or anonymised. It does not assume access to any University student database or integration interface. How student identity is verified,

which offices accept property, how long unclaimed property is kept and which notification channels are approved are open items that the study will establish or seek approval for (Appendix VI). The prototype will be tested with synthetic or anonymised data in a controlled setting and will not be presented as a deployed production system.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews concepts and prior work relevant to a web-based lost and found property management system with AI-assisted analytics. It is organised around the themes that arise from the study objectives: lost-and-found procedures and custody; web-based item registers; claims, ownership verification and audit trails; AI-assisted matching and descriptive analytics; data protection and security; and related projects. It ends by stating the research gap and a conceptual framework.

2.2 Review Approach

The review is thematic. Sources include peer-reviewed and conference papers on lost-and-found systems, undergraduate project reports held in university repositories, documentation of the scikit-learn library, methodological papers on design science research, Uganda's data-protection legislation and web-security standards. Repository reports are treated as project evidence rather than peer-reviewed findings, and statements about them in this chapter reflect the abstracts and records consulted so far. The full texts will be read before detailed claims are relied upon, and the review will be extended using the University library and academic databases.

2.3 Lost-and-Found Procedures and Custody

Lost-and-found handling in institutions normally depends on local procedure: a person reports a loss to an office, a finder hands an item to staff, and the receiving office records and stores it. The Makerere identification-card project illustrates a common weakness in such arrangements: when found items are left at several receptions, owners cannot tell where to look (Mirembe et al., 2023). The Busitema University project at Nagongera reports data gathering through interviews, observation, questionnaires and focus groups and a system that records lost items, found items, claims and users, with senior university officers able to manage an escalation process for claims (Musitwa, 2026). These examples suggest that custody and claim handling, not only listing, are part of the problem domain.

Two features of custody matter for system design. First, there is a difference between knowing that an item has been found and the institution holding it, so the record should show where the item physically is and who accepted it. Second, custody may pass between offices or staff, so transfers need to be recorded. How these steps are done at Nkumba, including which school offices accept found property and whether finders must deliver items to a particular office, is a requirements question for this study.

2.4 Web-Based Item Registers

A web-based register stores reports in a relational database with defined fields, so that users can browse and search by category, date, location and status without visiting offices. Browser access removes the need to install software, while the server validates input, enforces permissions and stores records. Existing campus systems follow this pattern: SearchiT at Makerere provides viewing, uploading and searching of lost and found items on a mobile application with a MySQL database (Itaagi, 2023), and FindItHere at UTHM uses PHP and MySQL with an administrative dashboard and notifications for matched items (Murugiah & Mat Nawi, 2025). A register is only as reliable as its data, so accurate entry, clear statuses and administrative oversight are essential, along with input validation and access control because users submit content that is stored and later shown to others.

2.5 Claims, Ownership Verification and Audit Trails

Publishing an item's full description helps owners find it but also allows anyone to copy the description and claim it. A listing should therefore show enough for recognition, such as category, general description and approximate place and date, while distinctive proof-of-ownership details, such as marks, serial numbers or contents, stay private and are used by authorised staff during verification. Which evidence is acceptable and who decides are institutional decisions, not technical ones, and the University must define them.

Claim handling should assume that more than one person may claim the same item. Each claim and each decision should be preserved with reasons, and claim history should be visible only to authorised staff. A rejection is an administrative decision recorded against a claim, not an accusation, and should never be shown publicly. Records of who created a report, who accepted custody, who reviewed a claim and who released an item give accountability and support dispute resolution. Role-based access control restricts each function and each data item to the roles that need it (Rangappa, 2023); least privilege means that even administrators should not automatically see claimant evidence.

2.6 AI-Assisted Matching and Descriptive Analytics

Two people can describe the same object in different words, for example a “black mobile phone” and a “dark smartphone”. TF-IDF represents each description as a vector in which a term’s weight rises with its frequency in that description and falls with how common it is across all descriptions, and cosine similarity measures the angle between two such vectors (Scikit-learn developers, n.d.-a, n.d.-b; Manning et al., 2008). Ranking found reports by similarity to a lost report gives a transparent, inexpensive method that needs no large image dataset. UTHM’s FindItHere system takes a similar practical route, using category filtering and text-based matching rather than heavier techniques (Murugiah & Mat Nawi, 2025).

The method has clear limits. TF-IDF is lexical: it matches shared words, not meaning, so synonyms and vague descriptions reduce its usefulness. Structured fields such as category, colour, brand, approximate date and location can supplement the text, but any combined score must have defined, published weights so that its behaviour can be explained and evaluated. Where practical, the interface should show why a pair was suggested, such as the matched terms and attributes. Image-based matching exists in other systems but is excluded here because it needs image data, raises additional privacy questions and is not required to evaluate the core workflow.

A similarity score expresses textual similarity between two records. It is not evidence that a claimant owns an item, and it must never approve release. People can describe items inaccurately, use common descriptions or make false reports, so candidate matches are suggestions for a custodian, who verifies a claim using private details and the University’s procedure.

Descriptive analytics summarise historical records: counts of reports by category, school or location, period and status; numbers of claims and returned items; and records awaiting custody. The database calculates exact counts and rates, and each figure should state its denominator and data period. Such summaries describe past usage and are not predictions. Generative text summaries are not part of the core design; if they were added later they would have to be grounded in database-calculated values and reviewed by a person. The NIST AI Risk Management Framework offers voluntary guidance for identifying and managing AI risks (National Institute of Standards and Technology, 2023); it is a reference for good practice, not Ugandan law.

2.7 Data Protection and Security

A lost-and-found system holds student identities, contact details, photographs if uploads are allowed, claim evidence and records of who handled what. In Uganda the Data Protection and Privacy Act, 2019 and the Data Protection and Privacy Regulations, 2021

are the primary legal framework for such personal data (Uganda, 2019, 2021), and the Personal Data Protection Office publishes registration and guidance notes for data controllers and processors (Personal Data Protection Office, n.d.). Whether and how these obligations apply to the University's deployment of a system like this, who is the controller, how long each category of data is kept, who may access it and whether any data is processed outside Uganda are questions to settle with the University; this proposal treats them as open items (Appendix VI) and does not assert legal conclusions. Foreign regulations are not applicable law here and are not used as such.

The design applies data minimisation, authentication, role-based authorisation, least privilege, server-side validation, secure password hashing, protected handling of any uploaded files and audit logging. The OWASP Top 10:2025 lists categories of web-application risk that map directly onto these controls, including Broken Access Control, Cryptographic Failures, Injection, Authentication Failures and Security Logging & Alerting Failures (OWASP Foundation, 2025). The OWASP Application Security Verification Standard (version 5.0.0) provides a basis for writing testable security requirements (OWASP Foundation, n.d.). These sources inform the security checklist used in design and testing in Chapter Three.

2.8 Related Projects

Several lost-and-found systems have been developed for universities in Uganda and elsewhere. **Table 2.1** summarises those consulted. Entries describe what the cited records and abstracts report; full texts will be read before detailed comparisons are relied upon.

Table 2.1: Related lost-and-found projects

Source	Context	What the record reports	Relevance and limitation for this study
Itaagi (2023), SearchiT	Makerere University, Uganda	Mobile application (Flutter, MySQL); viewing, uploading and searching items; claiming by email or phone; prototyping; unit, integration and user acceptance testing	Shows a local student-built register; claim by direct contact, with no custody or verification workflow described in the abstract

Source	Context	What the record reports	Relevance and limitation for this study
Mirembe et al. (2023)	Makerere University, Uganda	Online ID-card recovery system; students search using a photo of their own card; found cards otherwise left at various receptions	Documents dispersed found-item handling; limited to ID cards
Musitwa (2026)	Busitema University, Nagongera campus, Uganda	PHP and MySQL web tool built with RAD; records lost and found items, claims and users; officers manage an escalation process for claims	Closest local web system; focuses on one campus, and its abstract does not describe audit trails or AI matching
Murugiah & Mat Nawi (2025), FindItHere	UTHM, Malaysia	PHP and MySQL web application; category filtering and text-based matching; real-time notifications; administrative dashboard; Agile	Comparable text-matching approach; context and roles differ from Nkumba
Tan & Chong (2023)	University campus system	Study of a lost-and-found system for a university campus	Full text to be consulted to extract features and evaluation approach
David et al. (2024)	Three academic institutions	Comparative study of lost-and-found management systems, assessing reliability	Full text to be consulted for feature comparison

These sources show that lost-and-found systems for universities exist in Uganda and internationally, and this study does not claim otherwise. They also show common elements: item registers, search, claims and administrative functions, and in some cases matching and notifications.

2.9 Research Gap

Based on the records consulted so far, the related projects concentrate on listing, searching and claiming, with matching in some cases. The abstracts reviewed do not describe, in one system, all of the following: a clear separation between a posted found report and confirmed physical custody with school-level custody transfers; role-restricted claim review that preserves competing claims and decisions; moderation by soft deletion with a full audit trail; listings that withhold ownership evidence to resist false claims; and explainable candidate matching evaluated on a labelled test set. This statement is provisional until the full texts are read.

The gap this study addresses is therefore specific rather than general: to establish Nkumba University's actual requirements for custody, verification and privacy, to integrate these controls with campus-wide listings and bounded AI-assisted matching in one prototype, and to evaluate the result. It does not claim that no comparable system exists.

2.10 Conceptual Framework

The framework shows how inputs are processed into outputs that support lost-and-found management, with contextual factors that may affect the outcome.

Figure 2.1: Conceptual framework for the proposed system

Inputs	Processing	Outputs	Outcomes
Verified student accounts; lost reports; found reports; custodian receipt and custody records; claims and review decisions	Validation; role-based authorisation; custody and claim workflow; TF-IDF similarity ranking; audit logging; aggregation by the database	Approved listings by category; ranked candidate matches; custody and claim status; audit records; aggregate reports	Better organised records; easier discovery of found items; accountable verification and handover; support for recovery of property

Contextual factors: access to the system, data quality, user trust, custodian procedures and privacy safeguards. **Controls:** authentication, least-privilege roles, human verification of every claim and release.

The outcomes are expected, not assumed, and will be assessed through the evaluation in Chapter Three.

2.11 Chapter Summary

The review shows that lost-and-found handling depends on custody, claim verification and controlled information as much as on listing. Existing university systems provide registers, search and claims, with simple matching in some, but the records consulted do not describe the combination of custody separation, controlled claim review, auditable moderation and bounded evaluated matching proposed here. TF-IDF matching is a suitable explainable starting point if treated as a suggestion tool. Uganda's data-protection law and current web-security guidance frame the privacy and security requirements.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter describes how the system will be investigated, designed, built and evaluated. It sets out the research design, the study site, population and sample, the instruments and procedures, the system requirements and workflow specification, the AI boundary, the privacy and security requirements, the architecture and development process, the evaluation design, data analysis, quality assurance, ethics and limitations. Unknown University procedures are treated as research topics and are not assumed.

3.2 Research Design

The study uses an applied artefact-development approach guided by Design Science Research, because its main output is a software artefact that addresses a practical problem (Hevner et al., 2004). The Design Science Research Methodology of Peffers et al. (2007) provides the stages, shown in **Table 3.1**. The study is a case study of Nkumba University and uses mixed methods: qualitative methods to understand procedures and requirements, and quantitative methods to summarise survey responses and test results (Creswell & Creswell, 2023). Within the design and development stage, the prototype is built in short iterative cycles with feedback from custodians and students.

Table 3.1: Design science stages applied to the study

Stage (Peffers et al., 2007)	Application in this study	Objective
Problem identification and motivation	Investigate current reporting, custody, search, claim and handover practice at Nkumba	1

Stage (Peffers et al., 2007)	Application in this study	Objective
Define objectives of a solution	Derive functional, privacy and security requirements and the permission matrix	1
Design and development	Models, database, interfaces, workflow and status model; build the prototype and the matching component	2, 3
Demonstration	Run the test scenarios on synthetic or anonymised data	2, 3
Evaluation	Test functionality, usability, security and privacy, auditability and match relevance	4
Communication	Document design, evaluation and limitations in the project report	4

3.3 Study Site

The study is conducted at Nkumba University, which operates through seven constituent schools (Nkumba University, n.d.-a) and describes campuses at Entebbe and Kampala (Nkumba University, n.d.-b). The campus or campuses covered, the school offices that accept found property and the designated custodian of each are to be confirmed with the University before data collection (Appendix VI). Permission will be sought from the relevant authority.

3.4 Study Population

The population comprises Nkumba University students, who are the primary users; School Administrators or other designated custodians who would receive and release property; and system or ICT administrators who would manage accounts and the system. Other staff will be included only if the requirements research identifies them as actual actors in the process. The existing reporting tools, forms or registers will be reviewed where access is authorised.

3.5 Sample and Sampling

The final sample will be agreed with the supervisor once the accessible population and the University's approval are known. **Table 3.2** gives a provisional planning estimate of about 30–40 participants. It is not a confirmed figure and will not be presented as one.

Table 3.2: Provisional sample (planning estimate)

Group	Provisional size	Sampling technique	Role in the study
Students	20–25	Quota convenience sampling across schools	Requirements questionnaire; prototype testing and usability feedback
School Administrators and designated custodians	5–10	Purposive	Interviews; testing of custody, claim and handover functions
System or ICT administrators	2–5	Purposive	Interviews; testing of account, moderation and audit functions
Total (provisional)	**30–40**		

Purposive sampling is used for staff because they hold role-specific knowledge, and if the accessible staff population is small, all relevant staff may be included. For students, a probability sampling frame may not be available, so convenience sampling with quotas across the seven schools is proposed to avoid drawing all students from one school; its risk of selection bias will be reported as a limitation.

3.6 Data Collection Instruments

The following instruments are proposed. Their content is given in the appendices.

- **Student questionnaire** (Appendix I): current lost and found practice, discovery across schools, preferred categories and filters, expectations of account verification, found-item hand-in, claim and update preferences, privacy concerns and usability.
- **Custodian and staff interview guide** (Appendix II): which offices accept property, how receipt, storage and transfers are recorded, what student verification is authorised, how ownership evidence and competing or rejected claims are handled, who may edit or hide listings, retention rules, notification channels, access boundaries and desired reports.
- **Observation checklist** (Appendix III): used only with permission, recording process steps and information flow rather than personal data.
- **Document review checklist:** a tool for reviewing authorised forms, registers or written procedures without copying personal information.
- **Prototype test scenarios** (Section 3.12).

No instrument will ask participants for passwords, full identification numbers, item serial numbers or actual ownership proof.

3.7 Data Collection Procedures

Interviews. Semi-structured interviews with School Administrators, designated custodians and system administrators will establish how property is received, stored, transferred, claimed and released, and what rules and approvals apply. With permission, notes will be taken and summarised.

Questionnaires. Students will complete the questionnaire voluntarily, without giving names.

Observation. Where permitted, the researcher will observe the receipt, recording, searching, claim verification and handover steps, noting steps and information flow only.

Document review. Authorised forms, registers or written procedures will be reviewed. Personal information from any real record will not be copied into the research data unless it is necessary, authorised and protected; synthetic or anonymised test data will be preferred.

Prototype demonstration. After requirements analysis the prototype will be built and demonstrated with synthetic or anonymised records using the scenarios in Section 3.12.

3.8 System Requirements and Workflow Specification

3.8.1 Roles and permissions

The requirements are expressed as a permission matrix. The permissions below are proposed from the confirmed project requirements and will be adjusted to the University's approved procedures. One person may hold more than one role, and the system evaluates each role's permissions separately.

Table 3.3: Proposed actor, permission and control matrix

Actor	Permitted actions	Boundary and restrictions	Data recorded	Evidence (test)
Verified student	Browse all approved listings; search by category; submit own lost and found reports; submit and track own claims	Needs an approved student account; cannot see other students' private profile or claim evidence, or restricted storage details; cannot approve own claim	Report author, time, fields; claim attempts and status	Scenarios 1–6; role-boundary tests
School Administrator or custodian	Confirm physical receipt; record school custody and restricted storage reference; record custody transfers; review claims; record decision, reason and handover	Limited to assigned school or items unless University policy authorises wider access; all consequential actions logged	Receipt time, condition, accepting staff member, responsible office, storage reference; transfers; claim decisions; handover	Scenarios 4–10; audit review
System administrator	Manage accounts and roles; oversee listings; edit or hide, archive or withdraw with a reason; view approved aggregate analytics	No silent or permanent removal that bypasses accountability; claim evidence only when required by assigned duties	Account changes; edits and moderation actions with account, time, field or action, reason	Scenarios 11–12; security tests

3.8.2 Status model

A posted report must never be confused with a held item, so status is modelled in four separate dimensions rather than one field. The values below are the minimum and will be finalised with the University; statuses such as rejected, duplicate, withdrawn, archived or unclaimed are included only if the University approves them.

Table 3.4: Proposed status dimensions

Dimension	Possible values	Who can change it
Report publication	Submitted; approved and visible; hidden; archived; withdrawn	Author (withdraw); administrator or custodian (approve, hide, archive)
Physical custody (found items)	Reported found, awaiting physical delivery; received and accepted into custody; transferred to another office; released	Authorised School Administrator or custodian only
Claim state	No claim; submitted; under review; verified; rejected; withdrawn	Claimant (submit, withdraw); custodian (review, decide)
Item disposition	Held; approved for handover; returned or released; unclaimed or closed under University rules	Custodian; closure rules only as the University defines them

Rules the system must enforce: only an authorised custodian can confirm receipt into custody and record the responsible office and storage reference; more than one claim may exist against one item, and each claim and decision is preserved; a claimant cannot approve their own claim; a rejection is a recorded administrative decision and is never shown publicly; and hiding or archiving never deletes the record or its history. The exact state transitions will be defined after the University confirms its process.

3.8.3 Audit requirements

The system records, with account, timestamp and reason where applicable: who created a report; who accepted physical custody; each custody transfer; each claim attempt; each claim decision; each edit, including the changed field and, where feasible, the prior value for consequential changes; each hiding, archiving or withdrawal; and each handover or return.

3.9 AI Analytics Boundary

Matching. The core method is TF-IDF with cosine similarity over the text of item reports. Descriptions are pre-processed (case normalisation, punctuation removal, stop-word handling). Each lost report is compared with eligible found reports. A combined score may add category, colour, brand, approximate date and location agreement, but only with weights defined and documented before evaluation, and the text-only score will be reported separately for comparison. Output is a ranked list of candidate pairs with the matched terms and attributes shown where practical.

Inputs. Matching uses only the fields that are visible in approved listings. Private ownership clues, claim evidence, names, student numbers and contact details are not used.

Use of results. Candidate matches are suggestions for human review. They never prove ownership, never approve a claim and never trigger release. Who sees suggestions, and whether students see them, is an open item to settle with the University so that suggestions do not help a false claimant.

Analytics. Dashboard figures are exact database counts and rates: reports by category, school or location, date range and status; numbers of claims and returned items; and records awaiting custody. Each figure states its denominator and period.

Exclusions. The system does not assess people, claimant honesty, ownership or eligibility for release, and does not use facial recognition. No personal data or claim evidence is sent to an external AI service. If any external processing is later proposed, it must be disclosed and assessed under Uganda's data-protection requirements before inclusion.

3.10 Privacy and Security Requirements

Table 3.5: Proposed visibility of data (to be confirmed with the University)

Data item	Other authenticated users	Owner or author	Custodian (assigned school)	System administrator
Category, general description, approximate place and date, publication status	Visible	Visible	Visible	Visible

Data item	Other authenticated users	Owner or author	Custodian (assigned school)	System administrator
Student identity and contact details	Not visible	Own only	Only when needed for claim handling	Account management only
Private ownership details	Not visible	Own submission	Authorised review	Not by default
Claim evidence and claim history	Not visible	Own claims	Authorised review	Only when required by assigned duties
Restricted storage reference	Not visible	Not visible	Visible	As defined by duties
Audit history	Not visible	Not visible	Own school's items	Visible

The prototype will apply authentication, role-based authorisation with least privilege, server-side input validation, prepared database statements, secure password hashing, session protection, transport encryption in any deployment, protected handling of uploads if they are included (type and size checks, storage outside publicly accessible folders) and audit logging. These controls address the OWASP Top 10:2025 categories noted in Chapter Two (OWASP Foundation, 2025), and the OWASP Application Security Verification Standard will inform the wording of testable requirements (OWASP Foundation, n.d.). Student verification will use a University-approved process; the prototype does not assume integration with any student information system. Retention and deletion rules for accounts, claims, photographs and audit data will be defined with the University, and soft deletion must preserve accountability and access controls.

3.11 Architecture, Modules and Development Process

3.11.1 Architecture

The system uses a three-part web architecture. The presentation layer provides browser pages for students and staff. The application layer, written in PHP, validates requests, enforces role permissions, manages reports, custody, claims and audit records, and calls the matching component. The data layer is a MySQL database holding accounts, reports, custody records, claims, decisions and audit entries. The matching component, written in

Python with scikit-learn, receives only approved listing fields and returns ranked candidates to the application layer. How the application calls the matching component will be decided during design.

3.11.2 Main modules

- **Accounts and verification:** registration, student verification, login, roles and permissions.
- **Reports:** lost-item and found-item reports and their publication status.
- **Custody:** receipt confirmation, responsible office, storage reference and transfers.
- **Claims and verification:** claim submission, multiple claims per item, review, decision, reason and handover.
- **Moderation and audit:** edit, hide, archive and withdraw with reasons; complete action history.
- **Search:** browsing and filtering by category, date, location and status.
- **Matching:** pre-processing, TF-IDF, similarity ranking and explanations.
- **Analytics:** aggregate counts and rates for authorised users.

3.11.3 Development process

Work proceeds through requirements analysis, system design (use-case, activity or workflow, entity-relationship and architecture diagrams), iterative implementation, integration of the matching component, testing, and evaluation and refinement. Development uses a code editor, a local server, a database tool and version control; the tools may be adjusted with the supervisor.

3.12 Evaluation Design

Table 3.6: Evaluation criteria

Area	What will be checked	Evidence
Functionality	Whether each workflow in the scenarios works as specified	Test cases with pass or fail results
Usability	Whether selected users complete key tasks and understand the interface	Task completion observations; user feedback; optionally the System Usability Scale (Wafa et al., 2024)
Security and privacy	Whether role boundaries hold and private data stays private	Security test checklist and results

Area	What will be checked	Evidence
Auditability	Whether the history of any item can be reconstructed from the audit log	Audit review against test records
Match relevance	Whether known matches appear among the top suggestions	Precision at k, recall and error review
Analytics	Whether dashboard figures equal database counts	Comparison with direct queries on test data

Table 3.7: Functional test scenarios

No.	Scenario	Expected result
1	Account verification	Only a student who passes the approved check can post or claim
2	Browse and filter by category	Authenticated users see approved listings only
3	Lost report	Report is stored and visible without private details
4	Found report before hand-in	Status shows reported found, awaiting physical delivery; not held by the University
5	Custodian receipt	Only an authorised custodian can record custody, office and storage reference
6	Custody transfer	Transfer between offices is recorded with accounts and time
7	Valid claim	Claim is reviewed, verified, approved and handed over with a recorded reason
8	Rejected claim	Rejection is recorded and not visible publicly
9	Multiple competing claims	All claims and decisions are preserved and visible only to authorised staff
10	Approval and handover	Handover is recorded; AI score never approves release
11	Edit, hide or archive with reason	Listing leaves browsing but record and history remain
12	Audit review	Authorised user can trace every action on an item

Security and privacy tests will check role boundaries (for example, a student cannot open another student's claim evidence or a storage reference), hidden ownership details, input handling with valid and invalid data, session and password handling, and the completeness of the audit history.

Match evaluation. A labelled test set of lost and found pairs will be prepared from synthetic or anonymised descriptions. Its size, composition (categories, wording variety and look-alike items that are not matches), labelling method and limitations will be reported; the size will be agreed with the supervisor. Pairs will be labelled by the researcher and, where possible, checked by a second reviewer. Measures are precision at k, recall, and the share of known matches that appear in the top k suggestions, reported for text-only scoring, combined scoring and a simple keyword-overlap baseline. False positives and false negatives will be inspected. Accuracy will not be claimed from illustrative examples alone.

3.13 Data Analysis

Qualitative data. Interview notes, observation records and open questionnaire responses will be analysed thematically (Braun & Clarke, 2022). Codes will be grouped into themes such as reporting, custody, search, verification, handover, privacy and usability, and the themes will be used to derive and prioritise requirements.

Quantitative data. Questionnaire responses, task-completion results and test results will be summarised using frequencies, percentages and tables. No inferential claims will be made unless a justified design and sample support them.

Match and analytics data. Match results will be reported as precision at k, recall and the share of known matches in the top k suggestions, with an inspection of false positives and false negatives and a clear statement of the test set's size, composition and limits. Dashboard figures will be compared with direct database queries on test data.

Findings from the different sources will be compared to check whether they agree before requirements are finalised.

3.14 Quality Assurance

Instruments will be reviewed by the supervisor for clarity and fit with the objectives and tried with a few people who are not part of the final sample. Requirements will be traced to design components and to test cases so that every requirement is tested. The database will be checked for valid relationships and duplicate or inconsistent records. The prototype will undergo unit testing of individual functions, integration testing of module

interactions and system testing of complete workflows, with input validation tested using valid and invalid data. Security checks follow Section 3.12. For matching, the test pairs, labels and expected matches will be documented so that results can be reproduced.

3.15 Ethical Considerations

Permission and approvals. Permission will be sought from the relevant University authority before data collection, and any ethics or administrative approval required by the University or the School will be confirmed first.

Consent and voluntariness. Participants will be told the purpose of the study, what taking part involves, how their information will be used and that they may decline or withdraw without penalty.

Confidentiality and data minimisation. Only information necessary for the study will be collected. Questionnaires will not ask for names, passwords, full identification numbers, item serial numbers or ownership proof. Research records will be stored securely, accessed only by the researcher and authorised supervisors, kept only as long as needed and then deleted, in line with the retention rules to be confirmed.

Test data. The prototype will be tested with synthetic or anonymised data. No real student identities, claim evidence or ownership details from actual cases will be used.

Power relations. The researcher is a student at the case-study institution, and students may feel pressure when asked to take part by staff or peers. Participation will be voluntary, unconnected to any assessment, and staff will not be asked to comment on named students.

Honest reporting. Findings and test results will be reported as obtained; the proposed system will not be described as already developed, and benefits will not be claimed before they are measured.

3.16 Limitations of the Study

The prototype will be small in scale and tested in a controlled setting, so its behaviour under real, long-term use is unknown and production deployment is untested. The matching test set will be small and constructed, so results cannot be generalised to all types of property. The quality of matching depends on how well users describe items, and descriptions vary. Access to historical lost-and-found records may be limited. The student sample is a convenience sample and may be biased. Connectivity and hosting arrangements may limit demonstrations, in which case a local server will be used.

3.17 Chapter Summary

The study is a design-science case study of Nkumba University using mixed methods. Requirements will be gathered through questionnaires, interviews, observation and document review from a provisional sample of about 30–40 participants, and the prototype will be built iteratively with a permission matrix, a four-dimension status model, an audit trail and bounded TF-IDF matching. It will be evaluated against functional scenarios, security and privacy tests, auditability, usability and match-relevance measures, using synthetic or anonymised data and ethical safeguards.

REFERENCES

Braun, V., & Clarke, V. (2022). *Thematic analysis: A practical guide*. SAGE Publications.

Creswell, J. W., & Creswell, J. D. (2023). *Research design: Qualitative, quantitative, and mixed methods approaches* (6th ed.). SAGE Publications.

David, N. M., Catapang, G. M., Pabilando, K. P. V., Dala, L. T., Marcoso, R. F., & Lacasandile, A. D. (2024). A comparative study on lost and found management systems in academic institutions: Assessing reliability across Universiti Tunku Abdul Rahman, California State Polytechnic University, and Nazareth School of National University. In *2024 6th International Workshop on Artificial Intelligence and Education (WAIE)*. IEEE. https://doi.org/10.1109/WAIE63876.2024.00049

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly, 28*(1), 75–105. https://doi.org/10.2307/25148625

Itaagi, B., Jr. (2023). *A mobile application for finding lost items (SearchiT)* [Undergraduate dissertation, Makerere University]. Makerere University Dissertations. https://dissertations.mak.ac.ug/handle/20.500.12281/15582

Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*. Cambridge University Press. https://nlp.stanford.edu/IR-book/

Mirembe, R., Mwima, A. R., & Manzi, H. M. (2023). *Online university identification card recovery system for Makerere University* [Undergraduate dissertation, Makerere University]. Makerere University Dissertations. https://dissertations.mak.ac.ug/handle/20.500.12281/17480

Murugiah, V., & Mat Nawi, R. (2025). FindItHere: Lost and found item matching web application. *Applied Information Technology and Computer Science, 6*(2), 453–472. https://periodical.uthm.edu.my/index.php/aitcs/article/view/20420

Musitwa, A. F. (2026). *A web-based tool for reporting and recovering lost items for Busitema University students Nagongera* [Project report, Busitema University]. Busitema University Digital Repository. https://bdears.busitema.ac.ug/handle/123456789/9312

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). https://doi.org/10.6028/NIST.AI.100-1

Nkumba University. (n.d.-a). *NU profile*. https://nkumbauniversity.ac.ug/nu-profile/

Nkumba University. (n.d.-b). *Study at Nkumba University*. https://nkumbauniversity.ac.ug/study-at-nkumba-university/

OWASP Foundation. (n.d.). *OWASP application security verification standard* (Version 5.0.0). https://asvs.owasp.org/

OWASP Foundation. (2025). *OWASP Top 10:2025*. https://owasp.org/Top10/2025/

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Personal Data Protection Office. (n.d.). *Registration, classification and guidance notes*. https://pdpo.go.ug/media//2022/01/20102021105143-Registration_Classification_and_Guidance_Notes.pdf

Rangappa, A. S. (2023). An analytical study of role-based access control (RBAC) and attribute-based access control (ABAC) in database security for multi-tenant and cloud-based architectures. *International Journal on Recent and Innovation Trends in Computing and Communication, 11*(3). https://ijritcc.org/index.php/ijritcc/article/view/11943

Scikit-learn developers. (n.d.-a). *Pairwise metrics, affinities and kernels*. https://scikit-learn.org/stable/modules/metrics.html

Scikit-learn developers. (n.d.-b). *TfidfVectorizer*. https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

Tan, S. Y., & Chong, C. R. (2023). An effective lost and found system in university campus. *Journal of Information System and Technology Management, 8*(32), 99–112. https://doi.org/10.35631/JISTM.832007

Uganda. (2019). *Data Protection and Privacy Act, 2019* (Act No. 9 of 2019). https://ulii.org/en/akn/ug/act/2019/9/eng%402023-12-31

Uganda. (2021). *Data Protection and Privacy Regulations, 2021* (S.I. No. 21 of 2021). https://ulii.org/en/akn/ug/act/si/2021/21/eng%402021-03-12

Wafa, S., Maryadi, M. R., Mahendar, I. A., Fami, A., & Wahyoedi, B. (2024). Usability testing of the IPB student portal website using the System Usability Scale (SUS) method. *JATISI: Jurnal Teknik Informatika dan Sistem Informasi, 11*(3). https://doi.org/10.35957/jatisi.v11i3.8097

APPENDICES

APPENDIX I: STUDENT QUESTIONNAIRE

Introduction: This questionnaire collects requirements for a proposed web-based Lost and Found Property Management System at Nkumba University. Participation is voluntary and you may stop at any time. Please do not write your name, password, ID number, item serial numbers or any proof of ownership.

1. Which school are you in, and what is your year of study?
2. Have you lost property on campus? (Yes / No)
3. Have you found property on campus? (Yes / No)
4. If you lost or found something, how did you report or search for it? (School office / Notice board / Messaging group / Word of mouth / Other)
5. If you found an item, what did you do with it? (Handed it to a school office / Left it where found / Kept it to find the owner / Other)
6. Do you know which office accepts found items for your school? (Yes / No / Not sure)
7. What difficulties have you had in reporting or looking for lost property?
8. Would a campus-wide listing of found items, visible to Nkumba students and staff after login, be useful to you? (Yes / No / Not sure)
9. Which categories and filters would help you most? (Category / Date / Place / Status / Keyword)
10. How would you expect to be verified as a student when creating an account, and what would you find acceptable?
11. If you found an item, would you be willing to take it to the responsible school office after posting? What would make that easier?
12. How would you prefer to receive updates about a report or claim? (In-system / Email / SMS / Other)
13. What concerns do you have about your personal details or item details being shown in such a system?
14. What other features would you recommend?

Evaluation stage only: After using the prototype, rate each statement from 1 (strongly disagree) to 5 (strongly agree): I could create an account easily; I could find a listing easily; I could submit a report easily; I understood the status of my report or claim; I feel my personal information was protected.

APPENDIX II: CUSTODIAN AND STAFF INTERVIEW GUIDE

1. Which offices accept found property, and who is the designated custodian in each school?
2. If a student finds an item, what are they expected to do, and how should they know which office to use?
3. What do you record when an item is received (for example time, condition, storage reference, receiving staff member)?
4. How are items stored, who can access storage, and how are transfers between staff or schools recorded?
5. How can a student's identity be verified for an account, and what process or source is authorised?
6. What evidence of ownership is acceptable, who decides a claim, and what happens when several people claim one item or the evidence is inconclusive?
7. How should rejected claims be recorded and who may see them?
8. Who may edit, hide or remove a listing, and what reasons and records are required?
9. What rules apply to unclaimed property, including how long it is kept and what happens afterwards?
10. Can finder and claimant contact be arranged through the system rather than directly?
11. What are the rules for keeping and deleting student details, claims, photographs and audit records, and who may see each?
12. Which notification channels (email, SMS, in-system) are available and approved?
13. Are images necessary for the first version?
14. What reports or statistics would help you manage lost and found property?
15. What privacy or security concerns should the system address, and who should have access to what?

APPENDIX III: OBSERVATION CHECKLIST

Observe only with permission. Record steps and information flow, not unnecessary personal data.

1. How a found item is brought to the office and who receives it.
2. What is recorded at receipt and where.
3. How and where the item is stored and who can access the storage.
4. How a person enquiring about a lost item is assisted.
5. How an item is compared with an enquiry.
6. How a claimant is verified and what evidence is requested.

7. How a decision is made and recorded.
8. How the item is handed over and what record is kept.
9. How competing claims or disputes are handled.
10. How items are transferred between offices or staff.

APPENDIX IV: INDICATIVE WORK PLAN

Provisional planning estimate. Replace with dates and durations approved by the supervisor.

Activity	Weeks 1–2	Weeks 3–4	Weeks 5–6	Weeks 7–8
Review literature; refine proposal; obtain permissions	X			
Collect requirements	X	X		
System analysis and design		X		
Implementation and integration			X	
Testing and evaluation			X	X
Documentation and report				X

APPENDIX V: INDICATIVE BUDGET (UGX)

Provisional estimates. No hosting or deployment commitment is made unless the University approves it.

Item	Estimated cost (UGX)	Purpose
Internet and data	150,000	Research, development and testing
Transport	100,000	Visits to the University offices
Stationery and printing	80,000	Questionnaires and drafts
Hosting or domain (if approved)	100,000	Prototype deployment
Contingency	70,000	Unforeseen expenses

Item	Estimated cost (UGX)	Purpose
Total	**500,000**	

APPENDIX VI: REGISTER OF OPEN ITEMS TO CONFIRM

Working appendix. Resolve each item through approved requirements gathering, then remove this appendix before submission.

No.	Open item	Where it affects this proposal	Confirm with
1	Which school offices accept found property and who is the custodian of each	1.5, 3.3, 3.8	University, school offices
2	Must students deliver found items to the relevant office, and how does the system direct a finder who does not know it	3.8, Appendix I and II	School offices
3	What the custodian records on acceptance	3.8.1, 3.8.3	Custodians
4	Authorised student identity source or process; no assumed integration	1.5, 3.10	University, ICT
5	Acceptable ownership evidence, who decides, how competing or inconclusive claims are resolved	2.5, 3.8	Custodians, University
6	Whether finder and claimant contact is mediated by the system	3.8, 3.10	University
7	Custody transfer procedure, storage access and required audit records	3.8.1, 3.8.3	Custodians
8	Retention period and final disposition of unclaimed property	3.8.2	University

No.	Open item	Where it affects this proposal	Confirm with
9	Retention, deletion and access rules for accounts, claims, photos and audit data; any processing outside Uganda	2.7, 3.10	University, data-protection advice
10	Whether listings are visible to authenticated Nkumba users only (current default) or the public internet	1.5.1, 3.10	University
11	Notification channels available and approved	Appendix I and II	University, ICT
12	Whether images are needed in the first prototype	1.5.4, 3.9	Supervisor, University
13	Accessible populations, approved sample size, dates, budget and deployment environment	3.5, Appendices IV and V	Supervisor, University
14	Which campus or campuses are covered	1.5.2, 3.3	Supervisor, University
15	Whether listings need an approval step, who approves, and whether a found report is visible while awaiting delivery	3.8.2	University
16	Whether staff may post or claim, or only browse	1.5.1, 3.8.1	University
17	Who sees candidate-match suggestions and whether students do	3.9	University, supervisor